

A novel adaptive weighted fractional-order reverse accumulation grey model for predicting state-of-health in lithium-ion batteries

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ABSTRACT

Accurate prediction of lithium-ion battery state-of-health is essential for the safety, reliability, and lifetime optimization of energy storage systems, yet it remains challenging due to non-linear degradation dynamics, abrupt knee-point transitions, and limited historical data. Conventional grey prediction models are data-efficient but often fail to sufficiently emphasize recent observations, which are most informative for forecasting imminent degradation. To address this limitation, this study proposes a novel adaptively weighted fractional-order reverse accumulation grey model (AWFRAGM) that systematically strengthens the New Information Priority principle through a multi-layer synergistic architecture. Specifically, recent degradation information is amplified at the data level via fractional-order reverse accumulation, dynamically reweighted at the structural level using adaptive cycle-dependent weights, and further aligned with late-life degradation behavior through recency-aware metaheuristic optimization. Validation using the NASA B0005 and B0018 battery datasets demonstrates that AWFRAGM achieves testing mean absolute percentage errors of 0.53 % and 0.61 %, corresponding to an approximate 40–60 % reduction in prediction error compared with advanced grey models and hybrid neural network benchmarks. Statistical Diebold–Mariano tests confirm that these improvements are significant at the 5 % level. The results show that the proposed multi-layer reinforcement of recent information enables accurate tracking of degradation trajectories and enhanced responsiveness to knee-point transitions. Overall, the study provides a computationally efficient and practically deployable prognostic framework that advances grey system theory and supports next-generation battery management systems for future energy storage applications.

1. Introduction

1.1. Background

Lithium-ion batteries (LIBs) have become an essential technology driving the global transition towards electrified, low-carbon energy systems (Wu et al., 2025). Owing to their high energy density, long cycle life, and low self-discharge rate, LIBs dominate a wide range of applications, including electric vehicles (EVs), portable electronic devices, and grid-scale energy storage systems that support the integration of renewable energy sources (Dai et al., 2024; Ghosh, 2025; Wang et al., 2024). The rapid deployment of EVs and the increasing penetration of intermittent renewables such as solar and wind power have driven

sustained growth in the LIB market over the past decade, positioning lithium-based storage as a critical enabler of decarbonisation, energy security, and grid resilience (Nadimi and Goto, 2026).

Despite their technological maturity and widespread adoption, LIBs inevitably undergo performance degradation during repeated charge-discharge and storage cycles (Zeng et al., 2026). This degradation arises from a combination of complex and interrelated physicochemical mechanisms, including loss of cyclable lithium, active material degradation, growth of internal resistance, lithium plating, electrolyte decomposition, and mechanically induced electrode damage (Shen et al., 2026). As these processes accumulate, they lead to progressive capacity fade and power loss, adversely affecting operational efficiency, reliability, and safety (Qian et al., 2026). Of particular concern is the

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presence of a pronounced ageing knee-point, at which the degradation trajectory transitions from a relatively gradual and quasi-linear decline to a rapid and highly non-linear deterioration (Chen et al., 2025). This accelerated ageing phase is commonly associated with phenomena such as interfacial kinetic degradation (Hu et al., 2026), additive depletion (Sun et al., 2025), electrode saturation (Xu et al., 2025), and loss of electronic percolation pathways (Yang et al., 2025), and it significantly shortens the remaining service life of the battery. From a mathematical perspective, the knee-point may be operationally characterized as the cycle index at which the curvature of the capacity degradation trajectory increases significantly, marking the transition from quasi-linear ageing to accelerated nonlinear decay. If $C(q)$ denotes the discharge capacity at cycle q . The knee-point q^* can be approximated by examining the second-order variation (Eq. (1)) (Ni et al., 2025):

$$\Delta^2 C(q) = C(q+1) - 2C(q) + C(q-1) \quad (1)$$

where a sustained increase in $\Delta^2 C(q)$ indicates the onset of accelerated degradation. Equivalently, in continuous form, the knee-point corresponds to a marked increase in the magnitude of the second derivative $d^2 C(q)/dq^2$, reflecting a curvature change in the ageing trajectory. This transition signifies that degradation mechanisms begin to compound rather than accumulate linearly, thereby increasing safety risks and sharply reducing the remaining useful life. Accurately anticipating this high-curvature transition is therefore critical for reliable SOH prediction and motivates the development of recency-sensitive prognostic models.

The state-of-health (SOH) is the most widely used indicator for quantifying battery degradation and is typically defined as the ratio of the current usable capacity to the nominal (initial) capacity, expressed as a percentage (Q. Li et al., 2025; Lyu et al., 2026; Fu et al., 2026). Accurate SOH estimation plays a pivotal role in battery management systems (BMS), enabling reliable prediction of remaining useful life (RUL), optimisation of charge and discharge strategies, mitigation of safety risks (including thermal runaway), and informed decisions regarding battery retirement or second-life deployment (Mowri et al., 2025; Takiso and Yu, 2025). In practical applications, SOH assessment must be performed online using limited measurements (primarily voltage, current, and temperature) under highly variable operating conditions (Rengarajan and Anuradha, 2025; Yang et al., 2026).

However, as concluded by Shunli Wang et al. (2026), achieving accurate and robust SOH prediction remains a challenging task. Battery ageing is inherently non-linear, non-stationary, and strongly dependent on historical usage patterns, environmental conditions, and operational stressors such as temperature, current rate, and depth of discharge (Maheshwari et al., 2020). These challenges are further exacerbated in real-world scenarios by incomplete degradation data, measurement noise, and the need for computationally efficient algorithms suitable for embedded BMS implementation (Sun et al., 2026). Conventional SOH estimation approaches, including physics-based electrochemical models, data-driven learning methods, and hybrid techniques, often face difficulties in simultaneously capturing emerging degradation trends, maintaining robustness under small-sample conditions, and prioritising recent ageing information that is most indicative of imminent failure (Yaxuan Wang et al., 2025).

These limitations highlight the need for advanced modelling frameworks capable of adaptively emphasising recent degradation dynamics while preserving interpretability and computational efficiency. Such approaches are essential for improving the reliability of SOH prognostics, particularly in the late-life regime where rapid degradation occurs, and for supporting the safe and efficient operation of next-generation energy storage systems.

1.2. Motivations

Accurate prediction of SOH of LIBs is a prerequisite for the advancement of BMS, the assurance of operational safety, the

optimisation of energy utilisation, and the effective deployment of second-life battery applications in EVs and stationary energy storage systems. Although substantial progress has been made in SOH estimation and prognosis, several fundamental challenges remain unresolved, motivating the development of more robust, adaptive, and data-efficient modelling frameworks.

First, battery degradation is intrinsically non-stationary and highly non-linear (Maheshwari et al., 2020), driven by complex and interacting physicochemical processes such as loss of cyclable lithium, active material degradation, electrolyte decomposition, and lithium plating. A particularly critical feature of this degradation behaviour is the emergence of an ageing knee-point, beyond which capacity fade accelerates rapidly and safety risks increase significantly (Ni et al., 2025). Accurately anticipating this transition is essential for timely control actions and risk mitigation. However, many existing SOH prediction models struggle to capture such abrupt regime changes, especially under realistic operating conditions characterised by temperature variations, dynamic current profiles, and fluctuating depths of discharge (B. Wang et al., 2025; Chen et al., 2026). As a result, degradation severity is often underestimated, leading to delayed warnings and suboptimal battery utilisation strategies.

Second, SOH estimation in real-world applications is constrained by limited, noisy, and incomplete data (Chen et al., 2025). In online BMS implementations, full life-cycle information is rarely available, and sensor measurements may be affected by noise, drift, or irregular sampling. Data-driven approaches, while powerful, typically require large and representative training datasets and often exhibit limited generalisation capability across different battery chemistries, ageing paths, and usage patterns (Vanem and Wang, 2025; Yun et al., 2025). Physics-based and electrochemical models offer valuable interpretability but are computationally demanding and depend on internal parameters that are difficult to identify accurately in situ (Li et al., 2024; Sijing Wang et al., 2026). Hybrid methods attempt to bridge these paradigms but still face challenges in balancing prediction accuracy, model transparency, and real-time feasibility (Zhou et al., 2024; Jarraya et al., 2025).

Third, grey prediction models have attracted considerable attention for battery health forecasting due to their ability to operate effectively under small-sample, uncertain, and partially known conditions (Xu et al., 2023). Central to grey system theory (GST) is the principle of new information priority (NIP), which asserts that recent observations carry greater informational value for forecasting future system behaviour (Chen et al., 2016a; Yong Wang et al., 2025b). Despite this theoretical advantage, conventional grey models (GMs) and many of their extensions do not sufficiently emphasise recent degradation information when applied to battery ageing data (Chen et al., 2020; Li et al., 2025). In practice, these models may assign inadequate weight to the most recent charge–discharge cycles, precisely where early indicators of degradation acceleration and knee-points tend to emerge. Moreover, traditional accumulation operators can, under certain configurations, dilute the influence of new information and lack the flexibility required to accommodate the pronounced non-linearity and sudden rate changes that characterise LIB capacity degradation trajectories (Wang et al., 2024, 2023).

Importantly, in battery capacity sequences, recent observations are not merely temporally closer to the prediction horizon; they encode updated information about the currently dominant degradation mechanisms. As ageing progresses, electrochemical processes such as intensified side reactions, impedance growth, structural material degradation, and lithium plating may accelerate or become more pronounced, altering the instantaneous degradation rate. Consequently, the statistical characteristics of late-life data differ from those of early-life cycles, reflecting a shift in system dynamics. From this perspective, recent capacity measurements carry higher informational value because they better represent the present ageing regime and the near-future evolution of the system. Assigning insufficient weight to these observations may therefore dilute critical signals of degradation acceleration

and delay the detection of knee-point transitions.

These limitations collectively highlight the need for an advanced grey-based modelling framework that explicitly reinforces the NIP principle while enhancing modelling flexibility and adaptability. Such a framework should (i) adaptively prioritise recent degradation data through cycle-dependent weighting, (ii) employ fractional-order operators to better capture complex, non-linear ageing dynamics, (iii) utilise reverse accumulation to naturally amplify the contribution of new information during sequence construction, and (iv) incorporate efficient optimisation strategies to tune key model parameters under limited-data conditions.

Motivated by these considerations, this study proposes an adaptive weighted fractional-order reverse accumulation grey model (AWFRAGM), in which the fractional-order is optimised using the Greater Cane Rat Algorithm (GCRA) under a weighted error minimisation criterion (Agushaka et al., 2023). By explicitly aligning model structure and optimisation with the informational importance of recent degradation behaviour, the proposed approach aims to improve early detection of degradation acceleration, enhance SOH prediction accuracy, and provide a reliable and computationally efficient prognostic tool for practical BMS applications in next-generation energy storage systems.

1.3. Related studies and research gaps

Accurate prediction of LIB SOH has attracted sustained research interest due to its critical role in BMS, energy storage safety, and lifetime optimization. Existing studies on battery health estimation can be broadly classified into three main categories: statistical and feature-based models, intelligent data-driven approaches, and GST-based methods. Each category addresses specific aspects of battery degradation modelling, yet all encounter challenges when dealing with strong non-linearity, time-varying degradation rates, knee-point transitions, and limited historical data.

Statistical and feature-based models typically rely on extracting degradation-sensitive indicators from voltage, current, capacity, or impedance signals and establishing empirical relationships with battery health metrics. Rajan et al. (2018) employed probability density function frameworks combined with characteristic probability metrics and residual area capacity to construct linear or quasi-linear SOH predictors under varying charge–discharge conditions. Other studies have explored high-order polynomial fusion of multimodal features to capture complex ageing characteristics (Wang et al., 2025), dual-kernel Gaussian process regression for multi-feature SOH estimation (Richardson et al., 2018; Liu et al., 2019a, 2019b), and incremental capacity or differential voltage analysis to identify effective health indicators such as discharge slope evolution or peak shift (Zheng et al., 2018). Although these approaches offer physical interpretability and strong performance under controlled conditions, their accuracy is often sensitive to feature selection, operating regimes, and measurement noise, limiting their robustness across different battery chemistries and usage profiles.

Intelligent data-driven and deep learning-based approaches have also been extensively investigated for battery state estimation and prognostics. Neural networks, support vector machines, and recurrent architectures have demonstrated high predictive accuracy when trained on large, high-quality datasets (Chaoui and Ibe-Ekeocha, 2017; Xiong et al., 2020; Shu et al., 2021; Rao et al., 2024). In recent years, more sophisticated hybrid frameworks have been proposed to enhance robustness and adaptability. For example, Shunli Wang et al. (2025) developed a Bayesian optimization-based bidirectional LSTM model integrated with an unscented Kalman filter for high-precision state-of-charge estimation, achieving a maximum error of only 0.113 % under complex operating conditions. Similarly, S. Wang et al. (2023) proposed an improved anti-noise adaptive LSTM with a dual closed-loop observation strategy for remaining useful life prediction, significantly reducing RMSE, MAE, and MAPE while improving robustness under

varying temperatures and current rates. Furthermore, S. Wang et al. (2024) introduced a multiple feature–electrochemical thermal coupling model with real-time coefficient correction to address low-temperature performance degradation, demonstrating strong adaptability to current and temperature fluctuations.

While these intelligent and hybrid models achieve impressive accuracy, they generally require large volumes of high-quality training data, extensive hyperparameter tuning, and substantial computational resources. Their black-box nature also limits interpretability, and their performance may degrade under small-sample conditions or in online settings with incomplete historical information. These factors restrict their practical deployment in real-time BMS applications, particularly for early-stage or field-deployed energy storage systems where data availability is limited. In contrast, GST-based models provide a data-efficient alternative well suited to small-sample, uncertain, and partially known systems (Li et al., 2025). Classical GM(1,1) and its univariate extensions have been widely applied to battery capacity degradation and SOH prediction due to their simplicity and robustness. Early improvements focused on optimizing background values and whitening equations (Chen et al., 2016b), developing hybrid GMs to handle heterogeneous sequences (Chen et al., 2017, 2020), and introducing rolling or metabolic mechanisms to enhance adaptability (Chen et al., 2016b).

More recently, fractional-order GMs have emerged as a powerful extension of GST, offering enhanced flexibility to capture memory effects, non-integer dynamics, and the NIP principle (Sapnken et al., 2023; Yong Wang et al., 2025b; Wu and Zhang, 2018a). Adaptive and variable fractional-order accumulation operators have been proposed to better emphasise recent observations and accommodate evolving system trends (Xiong et al., 2011; Yong Wang et al., 2025c, 2025d). These approaches have demonstrated superior responsiveness in energy-related applications, including battery degradation modelling.

To further clarify the positioning of this work within recent advances, Table 1 summarizes representative studies on battery health estimation, highlighting their methodologies, data requirements, strengths, and limitations. This comparison reveals that existing high-accuracy approaches typically rely on large datasets and complex models, while grey-system-based methods remain data-efficient but insufficiently responsive to late-life degradation, motivating the proposed AWFRAGM framework. Despite these advances, existing grey-based SOH prediction models remain only partially aligned with the core degradation challenges discussed in Section 1.1, namely strong non-linearity, abrupt knee-point transitions, and small-sample constraints.

First, with respect to abrupt change and knee-point behaviour, standard first-order accumulation operators inherently smooth the original degradation sequence. While smoothing enhances noise resistance, it may simultaneously dilute emerging curvature changes and delay the detection of accelerated degradation phases (Yong Wang et al., 2025a). As a result, GMs based on fixed accumulation schemes may struggle to preserve the sharp inflection characteristics observed in late-life ageing. Second, regarding time-varying and nonlinear degradation rates, many existing GMs employ fixed background values or static weighting strategies (Sifeng, 2024). However, battery ageing is inherently cycle-dependent, with degradation velocity evolving over time. Fixed weights therefore fail to dynamically adjust to the increasing influence of recent observations, limiting responsiveness to nonlinear acceleration (Sapnken et al., 2026). This mismatch reduces predictive accuracy when degradation shifts from gradual to accelerated regimes. Third, under small-sample and recency-dominant conditions, parameter estimation is often conducted without explicitly incorporating temporal importance. Conventional optimization procedures treat all historical data uniformly, which may weaken the influence of late-life cycles that are more informative for forecasting imminent failure (Xie et al., 2021). Consequently, model calibration may not fully reflect degradation patterns near end-of-life.

Table 1

Comparative summary of representative studies on lithium-ion battery health prediction over the last decade.

Study	Methodology	Data requirement	Strengths	Main contribution	Key limitations
Chen et al. (2015)	Classical GM(1,1)	Low	Simple; data efficient	First application of GST to battery degradation	Poor adaptability; weak late-life accuracy
Chen et al. (2016b)	Optimized background GM	Low	Improved estimation stability	Structural refinement of GM	Limited response to abrupt degradation
Chaoui and Ibe-Ekeocha (2017)	Recurrent Neural Network	High	Captures temporal dependence	Early deep learning model for battery aging	Overfitting risk; black-box nature
Chen et al. (2017)	Hybrid GMs	Low–medium	Handles heterogeneous sequences	Expanded GM applicability	NIP not explicitly reinforced
Zheng et al. (2018)	Incremental capacity / DV analysis	High	Physically meaningful features	Identification of health indicators	Feature extraction sensitive to conditions
Richardson et al. (2018)	Dual-kernel GPR	High	Uncertainty-aware regression	Probabilistic SOH prediction	High computational cost
Liu et al. (2019b)	Gaussian Process Regression	High	Probabilistic output	Improved GP-based SOH estimation	Scalability issues
Xiong et al. (2020)	Support Vector Machine	Medium–high	Good generalization	ML-based SOH framework	Feature dependence
Chen et al. (2020)	Hybrid GMs	Low–medium	Robust to heterogeneity	Extended grey modelling	Insufficient late-life sensitivity
Shu et al. (2021)	ML ensemble	High	High accuracy	Ensemble learning for battery health	Data-hungry; limited interpretability
Wang et al. (2023)	Anti-noise adaptive LSTM	Very high	Robust to noise & parameter variation	Dual closed-loop RUL modelling	High complexity; large datasets
Wang et al. (2024)	Electrochemical–thermal coupling model	High	Handles low-temperature effects	Real-time coefficient correction	Complex modelling; limited small-sample use
Wang et al. (2024)	Fractional GM(1,N) + delay effect	Low	Handles dynamic delay between variables during charging and discharging	Better characterizes functions among variables with parallel relationships	Cannot handle data of nonlinear variation
Rengarajan and Anuradha (2025)	PDF-based statistical modelling	Medium–high	Physically interpretable	Statistical SOH framework	Sensitive to noise; limited non-linearity
Wang et al. (2025)	BO-BiLSTM-UKF fusion	Very high	Extremely high accuracy	Hyperparameter-optimized deep fusion	Data intensive; embedded deployment difficult
K. Li et al. (2025)	Hybrid GM + ensemble Kalman filter	Low	High accuracy; strong knee-point responsiveness	Takes into account capacity regeneration phenomenon	Cannot adapt to complex battery degradation laws
Sun et al. (2026)	Fractional DGM(1,N) polynomial model	Low	High accuracy and better stability	Introduces a dynamic information priority Fractional operator	Validation across more chemistries needed
This study – AWFragm	Adaptive weighting + fractional reverse accumulation + GCRA	Low	High accuracy; strong knee-point responsiveness	Multi-layer reinforcement of NIP principle	Validation across more chemistries needed

These limitations indicate that existing GMs, although data-efficient, do not systematically reinforce the NIP principle across accumulation, weighting, and parameter estimation stages. This gap motivates the development of a more adaptive grey modelling framework capable of (i) preserving nonlinear curvature changes, (ii) dynamically adjusting to time-varying degradation rates, and (iii) enhancing recency sensitivity under limited data conditions.

1.4. Contributions and novelty

This study addresses the limitations identified in existing SOH prediction approaches by developing an advanced univariate GM framework specifically designed for LIB degradation forecasting. The main contributions of this work are summarised as follows:

- i. **Adaptive reinforcement of the NIP principle:** An adaptive cycle-dependent weighting scheme is introduced to explicitly reinforce the NIP principle, ensuring that recent charge–discharge cycles exert greater influence during model optimisation. This mechanism enables the proposed model to dynamically emphasise emerging degradation information and improves sensitivity to non-linear transitions and accelerated ageing phases associated with knee-point behaviour.
- ii. **Fractional-order reverse accumulation for enhanced recency sensitivity:** A fractional-order reverse accumulation generating operator (ξ -FRAGO) is incorporated into the GM framework to naturally prioritise newer observations during sequence

construction. By combining reverse accumulation with fractional calculus, the model achieves greater flexibility in representing non-integer memory effects and abrupt changes in degradation rate, while remaining effective under small-sample and uncertain data conditions.

- iii. **Recency-aware metaheuristic optimisation of model parameters:** The fractional-order of the proposed model is optimised using a metaheuristic algorithm under a weighted error objective function derived from the adaptive NIP-based weights. This recency-aware optimisation strategy aligns parameter estimation with the most informative degradation stages, thereby improving prediction accuracy without increasing model complexity or data requirements.
- iv. **A fully adaptive and computationally efficient univariate grey structure:** The resulting adaptively weighted fractional-order reverse accumulation grey model (AWFRAGM) preserves the simplicity, interpretability, and low computational cost of classical univariate GMs, while significantly enhancing adaptability to evolving degradation dynamics. This makes the model suitable for real-time implementation within BMS.

The novelty of this work lies in the synergistic integration of adaptive NIP-based weighting, fractional-order reverse accumulation, and recency-weighted metaheuristic optimisation within a unified univariate GM framework for battery SOH prediction. To the best of the authors' knowledge, this combination has not been systematically explored in the existing battery health prognostics literature. Through

comprehensive validation on benchmark datasets, the proposed approach demonstrates superior capability in capturing late-life degradation uncertainty compared with conventional GMs, highlighting its potential for practical and reliable SOH prognostics.

1.5. Organization of this study

The remainder of this paper is organised as follows. Section 2 presents the methodological framework of the proposed AWFragM model, including the cycle-dependent weighting scheme, model formulation, parameter optimisation strategy, benchmarks, evaluation metrics, and statistical comparison methods. Section 3 describes the experimental validation, detailing the datasets and data sources, parameter settings, and additional tests conducted to assess model robustness. Section 4 discusses the empirical results, with particular emphasis on the predictive performance of the proposed AWFragM, comparative analysis against benchmarks, and the influence of the fractional-order on SOH prediction accuracy. Finally, Section 5 concludes the paper by summarising the main findings and outlining potential directions for future research.

2. Methods

The algorithmic framework of the proposed forecasting model is illustrated in Fig. 1. The pipeline is designed to prioritize recent degradation trends in LIBs, which are most indicative of future SOH. The process involves four core stages. First, cycle-specific weights are calculated based on the NIP principle, assigning greater importance to recent cycles; this explicitly overcomes the fixed or heuristic weighting schemes commonly adopted in existing GMs. Second, the AWFragM model is formulated by integrating reverse accumulation with fractional

calculus; in contrast to conventional accumulation operators, this step amplifies recent degradation signals and mitigates information dilution. Third, a metaheuristic algorithm optimizes the fractional-order ξ by minimizing a weighted error function, where the weights are derived from the first stage; this ensures that parameter estimation is aligned with late-life degradation behavior rather than being dominated by early-cycle data. Finally, the optimized AWFragM is validated against benchmarks using performance metrics and statistical significance tests.

2.1. Cycle-dependent weighting

Battery degradation is a non-stationary process; recent capacity fade trends are more predictive of imminent failure than older patterns from early life cycles (Wu and Zhang, 2018b). To embed this concept into the modeling framework, we employ the NIP principle (Cao et al., 2025), which asserts that observations closer to the present possess higher informational value for forecasting and should be assigned greater influence.

We quantify this principle using the matrix comparison approach to generate a weight for each cycle q in a dataset of m charge-discharge cycles (Yang et al., 2023). An $m \times m$ pairwise comparison matrix is constructed. For any pair of cycles (i, j) , a score of 1 is assigned to the more recent cycle and 0 to the earlier one, reflecting the higher relevance of newer data. The sum of scores across each row q yields its total score E_q , representing its cumulative “recency advantage”. The final weight Ω_q for the q -th cycle is obtained by normalizing these scores as shown in Eq. (2):

$$\Omega_q = \frac{E_q}{\sum_{j=1}^m E_j}, q = 1, 2, \dots, m. \quad (2)$$

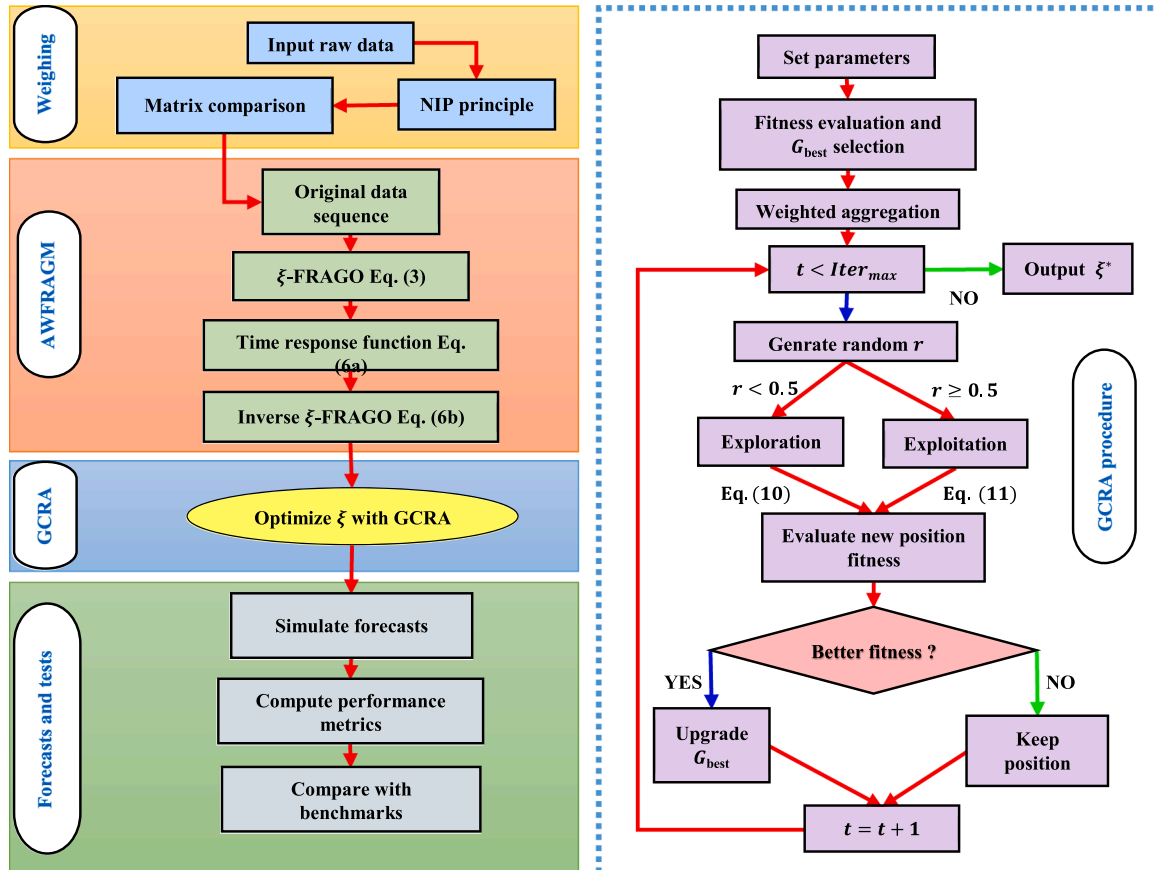


Fig. 1. Complete workflow of the proposed modelling framework.

These weights satisfy $\sum_{q=1}^m \Omega_q = 1$ and increase monotonically with q , ensuring the model's focus is adaptively biased toward the most recent degradation data. Compared with equal weighting, which implicitly assumes stationarity across the entire degradation trajectory, the proposed ordinal-comparison-based scheme explicitly encodes temporal asymmetry without imposing a predefined functional decay form. Unlike exponential decay weighting, which requires selecting a decay parameter and assumes a fixed rate of information attenuation, the matrix comparison approach introduces no additional hyper-parameters and relies solely on the relative temporal ordering of observations. This makes it scale-invariant and structurally robust under non-stationary conditions where degradation rates may evolve unpredictably. By enforcing monotonic recency prioritization without overly aggressive attenuation of intermediate cycles, the scheme preserves meaningful mid-life information while remaining highly responsive to abrupt late-life transitions, such as knee-point-induced acceleration. This weighting scheme makes the forecasting model inherently responsive to critical changes in the degradation rate, such as the onset of rapid knee-point fade, which is essential for accurate RUL prediction. From a grey system perspective, such order-based weighting aligns naturally with limited-sample conditions by avoiding over-parameterization while embedding structural recency bias.

2.2. Model formulation of the AWFRAGM

To prioritize recent degradation data, we adopt a reverse accumulation approach for sequence construction. This enhances the model's sensitivity to the latest trends in battery capacity fade, which are most critical for SOH estimation.

For an original battery capacity sequence $X^{(0)} = \{x^{(0)}(1), x^{(0)}(2), \dots, x^{(0)}(m)\}$, the traditional first-order accumulation generation operator (AGO) is defined as $x^{(1)}(p) = \sum_{i=1}^p x^{(0)}(i)$ (Sifeng, 2024). To increase model flexibility and accuracy, we introduce a fractional-order reverse accumulation generation operator (FRAGO). Let $Y = \{y_1, y_2, \dots, y_m\}$ represent a transformed sequence. Using the Gamma function $\Gamma(\cdot)$ to generalize the factorial, FRAGO is defined by Eq. (3):

$$y^{(\xi)}(q) = \sum_{i=1}^q \frac{\Gamma(\xi + q - i)}{\Gamma(q - i + 1) \Gamma(\xi)} y_i, \xi > 0, \quad (3)$$

where $y^{(\xi)}(q)$ is the ξ -order reverse-accumulated value (controlling memory depth) and q is the cycle index. Unlike forward accumulation, this operator inherently assigns higher effective weight to more recent observations when $\xi > 0$. The fractional-order introduces non-integer memory depth, allowing the model to flexibly balance historical smoothing and responsiveness to recent degradation.

From an engineering perspective, the reverse accumulation mechanism can be intuitively interpreted as modelling degradation from a present-state or RUL viewpoint. In practical battery management, control decisions are primarily based on the current health condition and its near-future evolution rather than distant historical behaviour. By constructing the accumulated sequence in reverse order, the model naturally emphasises the most recent degradation information, which better reflects the active ageing regime and the immediate risk of accelerated failure. This interpretation aligns the mathematical structure of the operator with the decision-oriented logic of real-time BMS implementations.

The core dynamic of the AWFRAGM is described by its whitening equation, a first-order differential equation governing the fractionally accumulated sequence defined by Eq. (4a) (Yong Wang et al., 2025d):

$$\frac{dy^{(\xi)}(t)}{dt} + \alpha y^{(\xi)}(t) = \beta \quad (4a)$$

To estimate the parameters α and β from discrete data, we discretize this equation. Integrating Eq. (4a) over the interval $t \in [q-1, q]$ and

applying the mean value theorem for integrals yields the fundamental discrete form of the model (Sifeng, 2024):

$$y^{(\xi)}(q) - y^{(\xi)}(q-1) + \alpha z^{(\xi)}(q) = \beta, q = 2, 3, \dots, m \quad (4b)$$

In Eq. (4b), $z^{(\xi)}(q) = 0.5(y^{(\xi)}(q) + y^{(\xi)}(q-1))$ is the background value. Rearranging Eq. (4b) gives a linear equation in α and β :

$$y^{(\xi)}(q) - y^{(\xi)}(q-1) = \beta - \alpha z^{(\xi)}(q) \quad (4c)$$

Writing Eq. (4c) for $q = 2, \dots, m$ generates a system of linear equations (S), which can be expressed in matrix form as $\mathbf{Y} = \mathbf{\Psi}\mathbf{P}$, where:

$$(S) : \begin{cases} \mathbf{\Psi} = \begin{pmatrix} -0.5(y^{(\xi)}(1) + y^{(\xi)}(2)) & 1 \\ -0.5(y^{(\xi)}(2) + y^{(\xi)}(3)) & 1 \\ \vdots & \vdots \\ -0.5(y^{(\xi)}(m-1) + y^{(\xi)}(m)) & 1 \end{pmatrix} \in \mathbb{R}^{(m-1) \times 2} \\ \mathbf{Y} = \begin{pmatrix} y^{(\xi)}(2) - y^{(\xi)}(1) \\ y^{(\xi)}(3) - y^{(\xi)}(2) \\ \vdots \\ y^{(\xi)}(m) - y^{(\xi)}(m-1) \end{pmatrix} \in \mathbb{R}^{(m-1) \times 1} \\ \mathbf{P} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \in \mathbb{R}^{2 \times 1} \end{cases} \quad (4d)$$

The least squares solution for the parameters $\hat{\mathbf{P}}$ is therefore:

$$\hat{\mathbf{P}} = \begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \end{pmatrix} = (\mathbf{\Psi}^T \mathbf{\Psi})^{-1} \mathbf{\Psi}^T \mathbf{Y} \quad (5)$$

Solving Eq. (4a) with the estimated parameters $\hat{\mathbf{P}}$ yields the time response function for the accumulated sequence given in Eq. (6a):

$$\hat{y}^{(\xi)}(q) = \left(y^{(\xi)}(1) - \frac{\hat{\beta}}{\hat{\alpha}} \right) e^{-\hat{\alpha}q} + \frac{\hat{\beta}}{\hat{\alpha}}, q = 1, 2, \dots \quad (6a)$$

The final SOH forecast on the original scale, $\hat{x}^{(0)}(k)$, is obtained by applying the inverse FRAGO to $\hat{y}^{(\xi)}(q)$. This inversion is implemented recursively:

$$\hat{x}^{(0)}(q) = \hat{y}^{(\xi)}(q) - \sum_{j=1}^{q-1} \frac{\Gamma(\xi + q - j)}{\Gamma(q - j + 1) \Gamma(\xi)} \hat{x}^{(0)}(j) \quad (6b)$$

with the initial condition $\hat{x}^{(0)}(1) = x^{(0)}(1)$.

Thus, the complete AWFRAGM forecasting procedure is: (i) transform data via Eq. (3), (ii) estimate α, β via Eq. (5), (iii) generate accumulated forecasts via Eq. (5a), and (iv) restore to the original scale via Eq. (6b).

2.3. GCRA-based optimization of the AWFRAGM

The value of ξ significantly influences the prediction accuracy of the AWFRAGM. To determine its optimal value, we employ the greater cane rat algorithm (GCRA), a swarm intelligence optimizer that effectively balances exploration and exploitation by simulating the dual-season foraging behavior of greater cane rats (Agushaka et al., 2023). The optimization minimizes the mean weighted relative error between predicted $\hat{y}^{(0)}$ and actual values $y^{(0)}$, where the weighting scheme from Section 3.1 emphasizes recent data. The objective function f_{obj} for optimization is defined as:

$$f_{obj}(\xi) = \min_{\xi} \left(\frac{1}{m-1} \sum_{q=2}^m \Omega(q) \frac{|y^{(0)}(q) - \hat{y}^{(0)}(q)|}{y^{(0)}(q)} \right) \quad (7)$$

In Eq. (7), $\Omega(q)$ are the adaptive cycle-dependent weights from Eq. (2). This weighting scheme ensures that prediction errors in recent

cycles contribute more strongly to the optimization process, aligning the optimization with the NIP principle. By embedding adaptive weights into $f_{obj}(\xi)$, the optimization stage consistently prioritizes recent degradation dynamics.

The GCRA optimization proceeds through the following six steps:

1. **Initialization.** Algorithm parameters are set as: population size $Pop = 50$, maximum iterations $Iter_{max} = 200$, and fractional-order bounds $\xi_{min} = 0$, $\xi_{max} = 2$. The search dimension is $\Theta = 1$ (as only ξ is optimized). The population is initialized uniformly within the bounds as in Eq. (8):

$$\xi_i = \text{rand} \times (\xi_{max} - \xi_{min}) + \xi_{min}, i = 1, \dots, Pop \quad (8)$$

For each candidate ξ_i , the fitness $f_{obj}(\xi_i)$ is evaluated using Eq. (7). The candidate with the smallest fitness is selected as the initial global best solution G_{best} .

2. **Fitness evaluation.** Evaluating $f_{obj}(\xi_i)$ requires executing the full AWFRAGM procedure: constructing the ξ_i -order reverse-accumulated sequence via Eq. (3), estimating coefficients α and β via Eqs. (4) to (5), generating fitted values via Eq. (6a), and computing the inverse accumulation to obtain predictions $\hat{y}^{(\xi)}(q)$.
3. **Weighted aggregation.** To concentrate the search, a weighted aggregation step is performed. The positions of some agents are updated toward G_{best} using Eq. (9):

$$x_i^{new} = \varpi \times \frac{x_i + x_q}{2} \quad (9)$$

where $\varpi = 0.7$ controls the aggregation strength.

4. **Exploration and exploitation phases.** The algorithm switches between exploration ($r < 0.5$) and exploitation ($r \geq 0.5$) phases. In the exploration phase, positions are updated relative to the dominant male's position x_q via Eq. (10):

$$x_i^{new} = x_i + \zeta \cdot (x_q - \xi_{food} \cdot x_i) \quad (10)$$

with acceptance governed by Eq. (11):

$$X_i = \begin{cases} \zeta \cdot (x_i - \alpha \cdot x_i) + x_i, & \text{if } f_{obj}(x_i) > f_{obj}(x_i^{new}), \\ \zeta \cdot (x_i - \beta \cdot x_i) + x_i, & \text{otherwise.} \end{cases} \quad (11)$$

In the exploitation phase, a randomly selected female individual n (with $n \neq q$) influences the update:

$$x_i^{new} = \zeta \cdot (x_q - \vartheta \cdot x_n) + x_i \quad (12)$$

In Eq. (12), ϑ is a stochastic integer parameter between 1 and 4. The new position is accepted only if it improves fitness.

5. **Iteration and termination.** After each iteration, fitness values are recalculated and G_{best} is updated if a superior solution is found. The process repeats until convergence or the maximum iteration limit is reached.
6. **Output.** The final G_{best} is returned as the optimized fractional-order ξ^* , which minimizes the weighted prediction error of the AWFRAGM.

2.4. Benchmark models

To rigorously evaluate the effectiveness of the proposed approach, the optimized AWFRAGM is compared against three benchmarks: an equally weighted variant of the proposed model known as EWFRAGM (Xie et al., 2021), a nonlinear discrete grey model optimized by ensemble Kalman filter (NDGM-EnKF) (K. Li et al., 2025), a back neural propagation neural network optimized by whale optimization algorithm (WOA-BPNN) (Che Wang et al., 2025).

EWFRAGM variant is identical to the proposed AWFRAGM in all

aspects except one: the weights $\Omega(q)$ in the optimization objective function (Eq. (7)) are set to be equal $(1/(m-1))$ for all cycles q . This comparison directly isolates the performance contribution of the NIP principle weighting scheme. The NDGM-EnKF model adopts the Ensemble Kalman Filter (EnKF) algorithm to dynamically correct and update the parameters of a Nonlinear Discrete Grey Model (NDGM) during the forecasting process. This approach not only prioritizes new data but also more fully utilizes the information in the data sequence, thereby improving forecasting stability and long-term prediction accuracy. The WOA-BPNN model is a hybrid neural network approach. A standard back-propagation neural network (BPNN) is used as the predictor, and its initial weights and biases are optimized globally by the Whale Optimization Algorithm (WOA) to avoid local minima and enhance convergence. This model serves as a representative of modern, data-driven machine learning techniques for time-series forecasting.

To ensure a fair and consistent comparison, all models (the proposed AWFRAGM and the three benchmarks) are applied to the identically preprocessed data, use the same training-testing split, and are evaluated using the same error metrics.

2.5. Evaluation criteria

Model prediction accuracy is assessed using two widely adopted error metrics: mean absolute error (MAE) and mean absolute percentage error (MAPE). These metrics provide complementary perspectives, measuring the absolute magnitude of errors and their relative size, respectively.

MAE is calculated by Eq. (13):

$$MAE = \frac{1}{m} \sum_{k=1}^m |x^{(0)}(q) - \hat{x}^{(0)}(q)| \quad (13)$$

MAPE is given by Eq. (14):

$$MAPE = \frac{100\%}{m} \sum_{q=1}^m \left| \frac{x^{(0)}(q) - \hat{x}^{(0)}(q)}{x^{(0)}(k)} \right| \quad (14)$$

The interpretation of the MAPE follows established criteria, as summarized in Table 2. A lower MAE or MAPE indicates superior forecasting accuracy. Together, these metrics offer a comprehensive assessment of model performance from both absolute and relative viewpoints.

2.6. Statistical comparison of model performance

Beyond point estimates of error, a statistical test is required to determine if the performance differences between models are significant. For this purpose, we employ the Diebold-Mariano (DM) test (Zhou et al., 2021), a formal hypothesis test designed to compare the predictive accuracy of two competing forecasting models.

Let $\{x_t\}$ be the actual time series, and let $\{\hat{x}_{it}\}$ for $i = 1, 2$ be the forecast sequences from two different models. The corresponding forecast errors are $e_{it} = x_t - \hat{x}_{it}$. The loss associated with a forecast at time t is defined by a loss function $\mathbb{L}(e_{it})$. A common choice is the squared error or absolute error. In this study, we use the MAPE at each point as the basis for the loss function to align with our primary accuracy metric. The

Table 2

Model accuracy classification based on MAPE range (Sapnken and Tamba, 2022).

MAPE	Accuracy rating	MAPE	Accuracy rating
Less than 5 %	Excellent	Between 10 % and 20 %	Moderate
Between 5 % and 10 %	Good	Above 20 %	Low

DM test evaluates the null hypothesis that the two models have equal forecast accuracy, i.e., that the expected difference in their loss functions is zero as in Eq. (15):

$$H_0 : \mathbb{E}[\mathbb{d}_t] = \mathbb{E}[\mathbb{g}(e_{1t}) - \mathbb{g}(e_{2t})] = 0 \quad (15)$$

If $\mathbb{E}[\mathbb{d}_t] < 0$, the second model is statistically superior; if $\mathbb{E}[\mathbb{d}_t] > 0$, the first model is superior.

The DM test statistic is constructed as follows. Let $\mathbb{d}_t = \mathbb{g}(e_{1t}) - \mathbb{g}(e_{2t})$ be the loss differential series. The statistic is calculated by Eq. (16):

$$DM = \frac{\chi - \frac{\kappa(\kappa+1)}{4}}{\sqrt{\frac{\kappa(\kappa+1)(2\kappa+1)}{24\kappa}}} \sim \mathcal{N}(0, 1) \quad (16)$$

where κ is the sample size and χ is the sum of signed ranks:

$$\chi = \sum_{t=1}^{\mu} I^+(\mathbb{d}_t) \cdot \text{rank}(|\mathbb{d}_t|) \quad (17)$$

In Eq. (17), $\text{rank}(|\mathbb{d}_t|)$ is the rank of the absolute loss differentials (from smallest to largest), and $I^+(\mathbb{d}_t)$ is an indicator function that equals 1 if $\mathbb{d}_t > 0$ and 0 otherwise. Under the null hypothesis, the DM statistic asymptotically follows a standard normal distribution.

In this study, the DM test is applied pairwise to compare the proposed AWFragM against each benchmark, using the MAPE-based loss differentials. A significant DM statistic (with a corresponding p-value < 0.05) allows us to reject the null hypothesis and conclude that one model's predictive accuracy is statistically superior to the other. The complete workflow of the proposed modelling framework, from data collection to forecasting evaluation is represented in Fig. 1.

3. Experimental validation and results

3.1. Dataset description and sources

To ensure objectivity, reproducibility, and comparability with existing studies, the experimental validation in this work is conducted using publicly available LIB ageing datasets that are widely adopted in the battery prognostics literature. Specifically, cycling data from batteries B0005 and B0018 provided by the NASA Ames Prognostics Center of Excellence ([Li-ion Battery Aging Datasets - NASA Open Data Portal WWW Document, n.d](#)) are employed. These datasets were obtained under controlled laboratory conditions and include long-term charge-discharge cycling records with measurements of capacity, voltage, current, and temperature. Owing to their completeness, stability, and extensive prior use, batteries B0005 and B0018 are considered highly representative benchmarks for evaluating SOH prediction models and have become reference datasets in LIB degradation studies. The degradation capacity of these two batteries is illustrated in Fig. 2.

Beyond their established benchmark status, batteries B0005 and B0018 were selected because their degradation trajectories exhibit pronounced nonlinear behaviour with observable late-life acceleration. In particular, both cells demonstrate a gradual capacity decline during early-to-mid cycles followed by a less stringent degradation phase as cycling progresses, reflecting the high-curvature ageing regime associated with knee-point transitions. This non-stationary pattern provides a meaningful testbed for evaluating models designed to emphasize recent degradation dynamics. By contrast, some other NASA cells display comparatively smoother or more quasi-linear degradation profiles over substantial portions of their lifespan. While such datasets are useful for general validation, they pose a less stringent test of a model's ability to capture abrupt acceleration phenomena. The deliberate selection of B0005 and B0018 therefore aligns with the core objective of this study: assessing whether the proposed recency-sensitive AWFragM framework can effectively track and forecast nonlinear accelerated degradation under realistic ageing conditions.

For each battery, the available capacity degradation data are divided

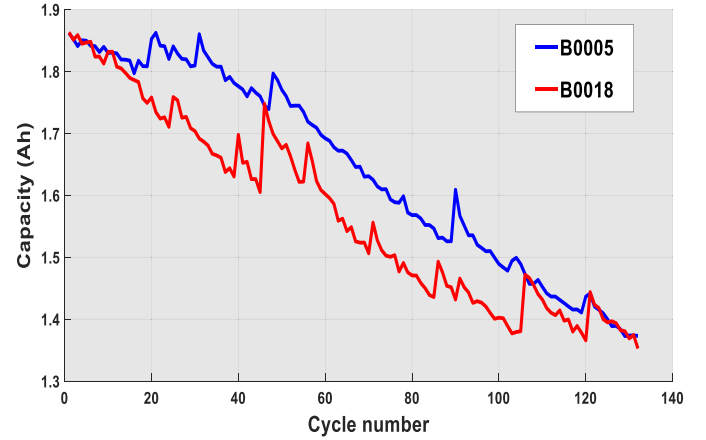


Fig. 2. Capacity degradation trajectories for NASA batteries B0005 and B0018. Measured SOH over full charge-discharge cycles shows characteristic non-linear fade.

chronologically into training and testing subsets using a 7:3 ratio. The training set (cycles 1–93) corresponds to the early-to-mid-life stage, during which capacity fade follows a relatively smooth and gradually evolving trend. This portion provides sufficient information for stable parameter identification and baseline trend learning. The testing set (cycles 94–132) represents the later-life region, where degradation curvature increases and accelerated fade or knee-point behaviour may emerge. This chronological split ensures that the model is evaluated on its capability to extrapolate from smooth early degradation into nonlinear accelerated ageing, which is more challenging and practically relevant than short-horizon interpolation. Such a setting reflects realistic BMS operation, where future SOH must be predicted using limited historical observations. Moreover, adopting identical training (93 cycles) and testing (39 cycles) horizons for both batteries ensures fairness and comparability across models, in line with common benchmarking practices in NASA battery degradation studies. The end-of-life (EOL) threshold is predefined as 70 % of the nominal capacity, which is widely used in LIB health assessment.

3.2. Parameter settings

This subsection details the selection and configuration of the key parameters for the proposed AWFragM model, its metaheuristic optimiser (GCRA), and the competing models. The rationale behind these settings is provided to ensure reproducibility and to justify the experimental design.

GCRA is used to optimize the fractional-order ξ of the AWFragM model under a weighted error objective function that emphasises recent degradation cycles, in accordance with the NIP principle. The algorithm parameters are set as follows: population size $Pop = 50$, maximum number of iterations $Iter_{max} = 200$, and fractional-order search bounds $\xi_{min} = 0$ and $\xi_{max} = 2$. The search interval $[0, 2]$ is adopted because fractional-order GMs typically operate within this range to preserve physical interpretability and numerical stability (Wu and Chen, 2025). The choices $Pop = 50$ and $Iter_{max} = 200$ follow commonly recommended settings in the original GCRA framework and were further validated through preliminary trials to balance convergence stability and computational efficiency (Agushaka et al., 2023). To assess robustness, a sensitivity analysis was conducted by varying Pop within $[30, 70]$ and $Iter_{max}$ within $[100, 300]$. The resulting variations in the optimal ξ remained below 2 %, and prediction accuracy (in terms of MAE and RMSE) changed by less than 3 %, indicating that the optimization outcome is stable with respect to these hyperparameters. The aggregation strength $\varpi = 0.7$ and stochastic parameters (ζ and θ) follow the standard configuration described in Section 2.3. Uniform random

initialization is adopted to ensure sufficient population diversity. Overall, these settings provide reliable convergence while maintaining moderate computational cost, supporting reproducibility and practical implementation.

For the EWFRAGM benchmark, the model is identical to the AWFRAGM except that the weights $\Omega(q)$ in the optimization objective function are set equally for all cycles: $\Omega(q) = 1/(m - 1)$, where m is the length of the training/testing sequence. This isolates the effect of adaptive weighting in the proposed model.

The NDGM-EnKF and WOA-BPNN benchmarks are implemented according to their standard definitions. For NDGM-EnKF, the Ensemble Kalman Filter is configured with an ensemble size of 50 to update NDGM parameters iteratively. For WOA-BPNN, the Whale Optimization Algorithm is run with a population of 30 for 100 iterations to optimize the neural network's initial weights and biases, while the BPNN architecture uses a single hidden layer with 10 neurons. All other parameters follow the default settings recommended in the respective literature. No additional fine-tuning is performed to ensure a fair and consistent comparison across models.

Tables 3 and 4 summarise the key parameter settings for the proposed AWFRAGM optimized by GCRA, and benchmark models, along with their MAEs and MAPEs performance. Execution time for training and testing phases (with results reported in Table 5) is measured using a standard computing environment to provide an indication of the computational efficiency of each approach.

Table 5 compares the average execution times of GCRA with six widely used metaheuristic optimizers on both battery datasets. For B0005 and B0018, GCRA consistently exhibits the lowest computational time, indicating faster convergence and superior computational efficiency. This advantage reflects the balanced exploration/exploitation mechanism of GCRA, which reduces redundant searches and accelerates parameter convergence. The results confirm that the improved predictive accuracy of AWFRAGM is achieved without incurring additional computational overhead.

3.3. Estimated SOH of lithium battery

This subsection reports the estimated SOH results obtained using the proposed AWFRAGM and competing models for batteries B0005 and B0018. The analysis includes model prediction accuracy, residual distributions, comparative performance against benchmarks, sensitivity to the fractional-order, and the consistency of estimated SOH trajectories with known physical degradation behaviour.

3.3.1. AWFRAGM performance

Figs. 3 and 4 present the estimated SOH trajectories generated by AWFRAGM over the full life cycles of batteries B0018 and B0005, respectively. For both datasets, the predicted SOH curves closely align with the measured capacity degradation during both the training and testing phases. AWFRAGM maintains a consistent monotonic decreasing

Table 3

Parameter settings for AWFRAGM and benchmarks, with comparative MAE and MAPE on training and testing datasets for battery B0005.

Model	Key parameters	MAE		MAPE	
		Train	Test	Train	Test
AWFRAGM-GCRA	$Pop = 50, Iter_{max} = 200, \xi \in [0, 2], \sigma = 0.7$	0.012	0.018	0.801	0.530
EWFRAGM	$\Omega_q = 0.00758, \forall q, \text{ default fractional-order}$	*	*	*	*
NDGM-EnKF	Ensemble size = 50, default NDGM orders	0.019	0.029	3.900	3.120
WOA-BPNN	WOA: $Pop = 30, Iter_{max} = 100$; BPNN: 1 hidden layer, 10 neurons	0.016	0.024	2.453	2.025
		0.028	0.041	4.136	5.698

* Indicates the best statistic

Table 4

Parameter settings for AWFRAGM and benchmarks, with comparative MAE and MAPE on training and testing datasets for battery B0018.

Model	Key parameters	MAE		MAPE	
		Train	Test	Train	Test
AWFRAGM + GCRA	$Pop = 50, Iter_{max} = 200, \xi \in [0, 2], \sigma = 0.7$	0.019	0.009	0.710	0.606
EWFRAGM	$\Omega_q = 0.00758, \forall q, \text{ default fractional-order}$	*	*	*	*
NDGM-EnKF	Ensemble size = 50, default NDGM orders	0.018	0.027	3.762	3.843
WOA-BPNN	WOA: $Pop = 30, Iter_{max} = 100$; BPNN: 1 hidden layer, 10 neurons	0.015	0.022	2.254	2.087
		0.027	0.039	4.005	5.208

* Indicates the best statistic

Table 5

Average execution time of metaheuristic algorithms. Performance comparison of the GCRA against six alternative optimizers on the two battery datasets.

Dataset	Optimizer						
	WOA	PSO	GA	FOA	GWO	FA	GCRA
B0005	5.27	4.15	6.31	4.48	5.02	6.87	3.42*
B0018	5.41	4.32	6.54	4.66	5.19	7.12	3.58*

* Indicates the best performing optimizer

trend and accurately tracks the overall evolution of battery ageing throughout the prediction horizon.

Quantitative accuracy results are summarised in Tables 3 and 4. For battery B0005, AWFRAGM achieves the lowest MAE and MAPE values among all considered models in both the training and testing phases. In particular, the testing MAE and MAPE obtained by AWFRAGM are substantially lower than those of EWFRAGM, NDGM-EnKF, and WOA-BPNN. A similar pattern is observed for battery B0018, where AWFRAGM again yields the smallest error metrics across all models and evaluation phases.

Compared with EWFRAGM, the proposed AWFRAGM consistently demonstrates reduced prediction errors, indicating improved SOH estimation accuracy when adaptive cycle-dependent weighting is employed. Relative to NDGM-EnKF, AWFRAGM exhibits lower MAE and MAPE values, particularly during the testing phase, where degradation dynamics become more pronounced. The data-driven WOA-BPNN model shows the largest prediction errors on both datasets, especially in the later cycles.

Across both batteries, AWFRAGM maintains stable prediction accuracy when transitioning from the training to the testing interval. The relative ranking of models based on MAE and MAPE remains consistent for B0005 and B0018, with AWFRAGM outperforming NDGM-EnKF, EWFRAGM, and WOA-BPNN in all evaluated cases.

3.3.2. Comparison with benchmarks

The comparison is conducted through an analysis of residual distributions, defined as the difference between the measured battery capacity and the corresponding model estimates. Figs. 5 and 6 illustrate the residual distributions obtained for batteries B0018 and B0005, respectively. For both datasets, the residuals produced by AWFRAGM are concentrated within a narrow band around zero across the entire life cycle, indicating small estimation errors during both the training and testing phases. All the residual values remain within $\pm 5\%$ range, with no pronounced error accumulation near the EOL region.

NDGM-EnKF model yields the second-best performance among the competing models. Its residual distributions are mostly concentrated in the $\pm 5\%$ band but exhibit a wider spread than those of AWFRAGM, particularly during the later degradation stages. A slight increase in residual dispersion is observed in Fig. 5a (near the testing horizon),

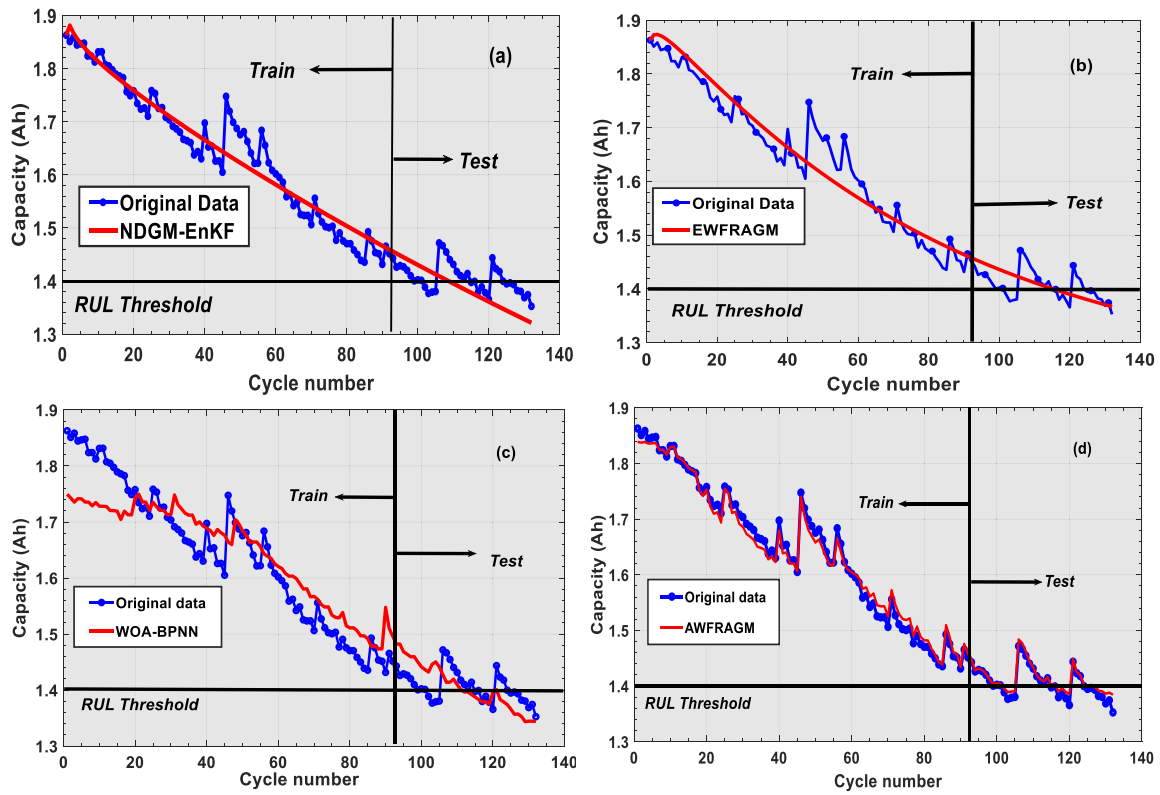


Fig. 3. SOH prediction results for battery B0018. Comparison of predicted versus measured SOH for the four models over the full lifecycle.

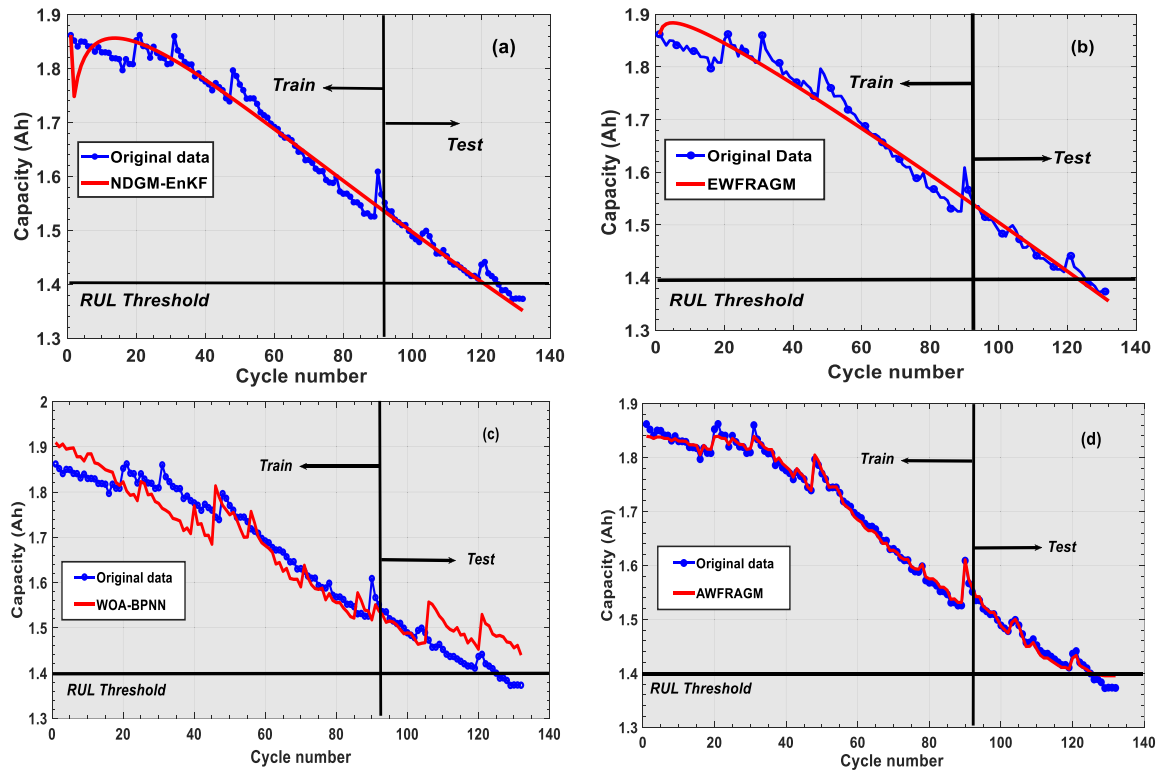


Fig. 4. SOH prediction results for battery B0005. Comparison of predicted versus measured SOH for the four models over the full lifecycle.

reflecting larger deviations in long-term SOH estimation.

EWFRAGM demonstrates relatively compact residuals during the early cycles; however, its residual distributions broaden during mid-life

and late-life stages. For both batteries, EWFRAGM displays heavier tails compared with AWFRAGM, with an increased frequency of larger errors in cycles approaching the EOL threshold.

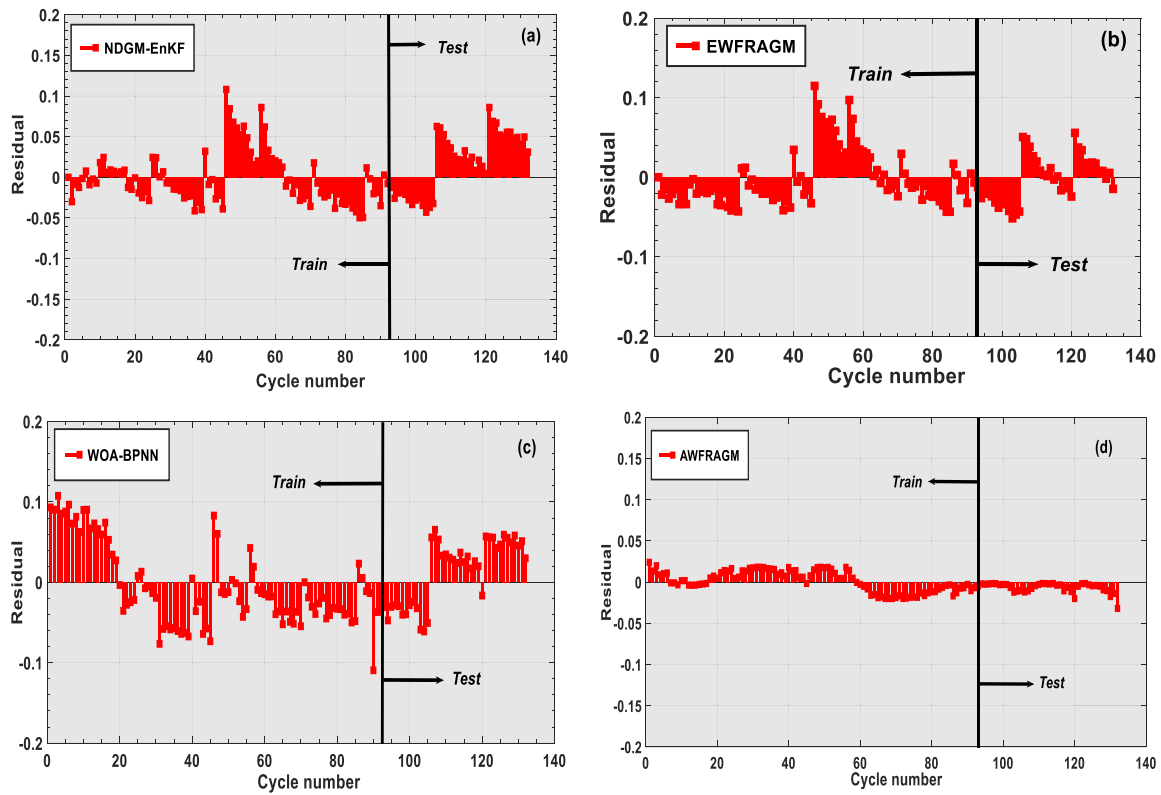


Fig. 5. Prediction error analysis for battery B0018. Residual distributions for the four models across all cycles.

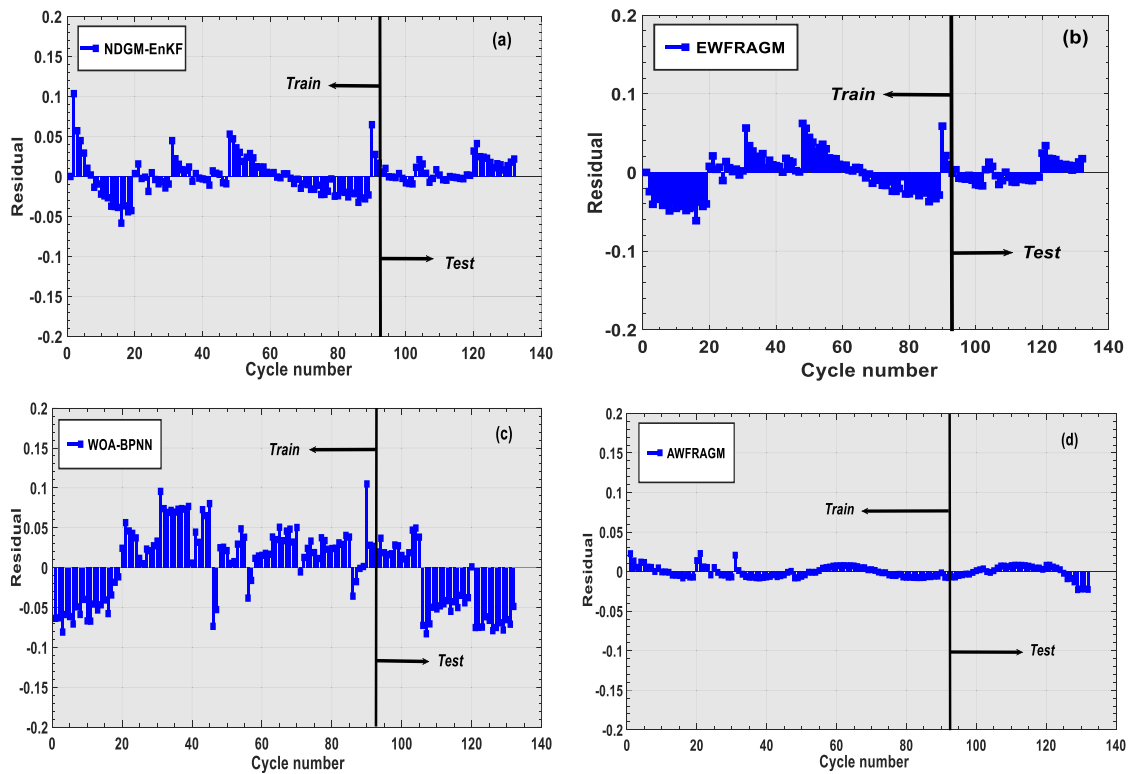


Fig. 6. Prediction error analysis for battery B0005. Residual distributions for the four models across all cycles.

The WOA-BPNN model produces the most dispersed residual distributions among all compared models. For both B0005 and B0018, the residuals exhibit wide variability throughout the testing phase, with

frequent large-magnitude deviations. Notably, error magnitudes approaching 10 % occur more frequently for WOA-BPNN, particularly under long-horizon prediction conditions.

Table 6 quantitatively confirms the superiority and stability of AWFRAGM for battery B0018. In the testing phase, which corresponds to late-life accelerated degradation, AWFRAGM achieves the lowest RMSE (0.012) and smallest residual dispersion ($SD = 0.011$), indicating strong predictive stability under nonlinear ageing conditions. Moreover, its residual skewness remains close to zero (0.09), suggesting minimal systematic bias. In contrast, benchmark models exhibit larger dispersion and increasingly positive skewness during testing, particularly WOA-BPNN (skewness = 1.04), reflecting asymmetric errors in the high-curvature degradation regime.

The residual analysis reported in Table 7 for battery B0005 further supports the robustness of AWFRAGM across different degradation patterns. During testing, AWFRAGM maintains the lowest RMSE (0.022) and smallest standard deviation (0.020), indicating controlled error dispersion even as ageing progresses. Its skewness remains very close to zero (0.14), confirming the absence of pronounced systematic bias. By comparison, the benchmark models show higher variability and markedly stronger positive skewness (especially WOA-BPNN (1.08)) suggesting less stable performance and asymmetric error accumulation in the later-life stage.

3.3.3. Sensitivity analysis

The sensitivity of the proposed AWFRAGM model to the fractional-order ξ is examined by varying ξ within the predefined search interval $[0, 2]$ while keeping all other parameters fixed. In order to provide a quantitative assessment, ξ is continuously swept across the interval, and the corresponding testing MAE and MAPE values are systematically recorded. For each value of ξ , SOH estimation results are generated for batteries B0005 and B0018 using the same training-testing partitions.

The resulting SOH trajectories and associated error metrics demonstrate a clear dependence on the fractional-order. Fig. 7 further illustrates the continuous error landscape over $\xi \in [0, 2]$, revealing a well-defined minimum region. The GCRA-optimized value ($\xi^* = 0.85$) is located at the bottom of this region, confirming that the metaheuristic search converges near the global performance optimum within the explored range. When lower fractional-orders ($\xi < 0.5$) are employed, the accumulated sequences exhibit reduced smoothing, leading to increased variability in the estimated SOH (see Fig. 7). Under these conditions, both MAE and MAPE values increase, and the corresponding APE distributions become wider, with a higher frequency of large-magnitude errors, particularly during the testing phase.

As the fractional-order increases, the estimated SOH trajectories become progressively smoother. For larger fractional-orders ($\xi > 1.5$), the SOH estimates display delayed responses to degradation changes, especially in the later cycles. This behaviour is reflected in elevated error levels near the EOL region and broader residual distributions compared with intermediate fractional-order settings.

The fractional-order ξ^* optimised via GCRA strategy consistently yields the lowest MAE and MAPE values for both batteries. At this optimal setting ($\xi^* = 0.85$), the APE distributions are the most concentrated around zero, with the narrowest spread across both training and testing phases (see Fig. 7). Moreover, the relatively flat curvature of the error curve near ξ^* indicates the presence of a robustness interval, within which moderate deviations of ξ result in only

Table 6
Residual statistics for training and testing phases (Battery B0018).

Model	Phase	RMSE	Std. Dev.	Skewness
AWFRAGM	Train	0.023	0.021	0.12
	Test	0.012	0.011	0.09
EWFAGM	Train	0.024	0.023	0.48
	Test	0.033	0.031	0.72
NDGM-EnKF	Train	0.019	0.018	0.36
	Test	0.027	0.025	0.58
WOA-BPNN	Train	0.032	0.030	0.81
	Test	0.045	0.042	1.04

Table 7

Residual statistics for training and testing phases (Battery B0005).

Model	Phase	RMSE	Std. Dev.	Skewness
AWFRAGM	Train	0.015	0.014	0.10
	Test	0.022	0.020	0.14
EWFAGM	Train	0.024	0.022	0.55
	Test	0.034	0.031	0.73
NDGM-EnKF	Train	0.020	0.019	0.39
	Test	0.028	0.026	0.60
WOA-BPNN	Train	0.033	0.031	0.84
	Test	0.047	0.044	1.08

marginal performance degradation. This property is particularly relevant for practical BMS applications where continuous re-optimization may not always be feasible.

These results indicate that SOH estimation accuracy varies significantly with the choice of fractional-order and that adaptive optimisation of ξ is essential for achieving stable and accurate performance across different degradation stages.

3.3.4. Convergence and stability analysis of GCRA

To verify the optimization stability of GCRA for fractional-order identification, the convergence behaviour of the algorithm is illustrated in Fig. 8. The objective function exhibits a rapid decrease during the early exploration phase, followed by gradual refinement in later iterations, indicating an effective balance between exploration and exploitation. Convergence is typically achieved well before reaching the maximum iteration limit, confirming computational efficiency.

To further assess robustness against stochastic effects, the optimization process was repeated over 20 independent runs with different random initializations. The results are reported in Table 8. The obtained optimal fractional-order parameter ξ shows a standard deviation below 1.5 %, while the corresponding variation in MAE remains under 2 %. These results demonstrate that the GCRA-based optimization yields stable and reproducible solutions, mitigating concerns regarding stochastic variability in metaheuristic search procedures.

3.3.5. Targeted analysis around the knee-point transition

To rigorously assess model responsiveness to degradation regime shifts, a localized evaluation was conducted within a symmetric ± 15 cycle window centred on the identified knee-point for both batteries B0005 and B0018. The knee-point was determined from the curvature change in the SOH trajectory, corresponding to the onset of accelerated capacity decay. The localized prediction errors are summarized in Table 9. As expected, all models exhibit higher MAE and MAPE values within this window compared to their overall testing performance, confirming that the knee-point region constitutes the most challenging forecasting phase.

For battery B0005, AWFRAGM achieves a MAE of 0.0078 and MAPE of 0.92 %, outperforming EWFAGM (0.0116), NDGM-EnKF (0.0139), and WOA-BPNN (0.0164). This corresponds to a MAE reduction of approximately 33 % relative to EWFAGM. For battery B0018, which exhibits a slightly more abrupt post-knee degradation, all models show increased errors. Nevertheless, AWFRAGM maintains the best localized performance (with 0.0091 MAE and 1.05 % MAPE), compared to EWFAGM (0.0128), NDGM-EnKF (0.0154), and WOA-BPNN (0.0182). The relative improvement remains consistent, with approximately 29 % MAE reduction over EWFAGM and nearly 50 % over WOA-BPNN.

Notably, NDGM-EnKF which employs a fixed accumulation structure demonstrates delayed adaptation to the sudden slope variation, resulting in systematic underestimation of accelerated degradation. In contrast, the adaptive weighting mechanism and fractional-order reverse accumulation structure of AWFRAGM enable faster alignment with evolving degradation dynamics. These results confirm that the superiority of AWFRAGM is particularly pronounced during the regime-transition phase, thereby directly substantiating the advantage in

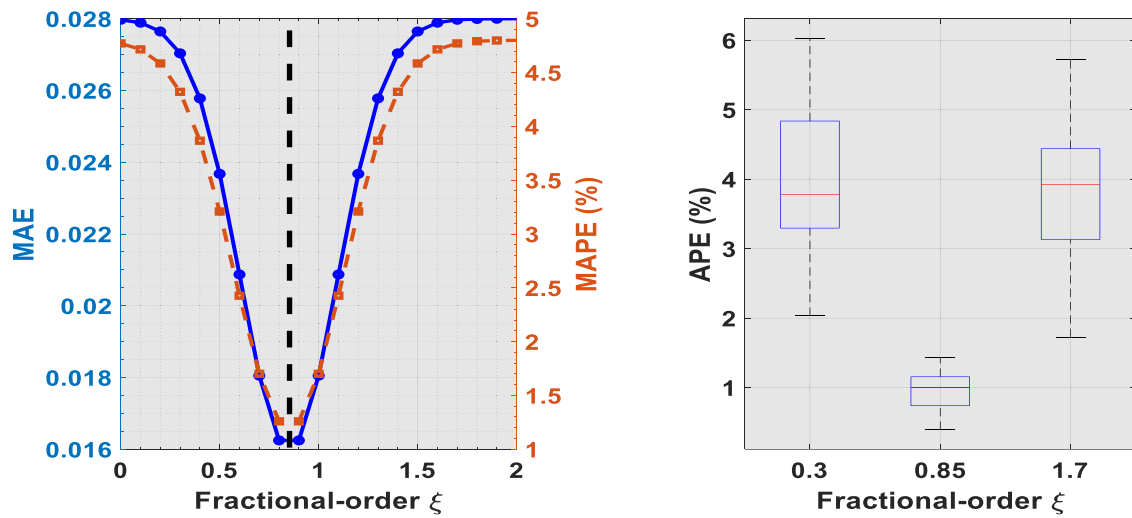


Fig. 7. Impact of fractional-order parameter on prediction accuracy. The dashed black line marks the GCRA-optimized value ($\xi^* = 0.85$) corresponding to the minimum error.

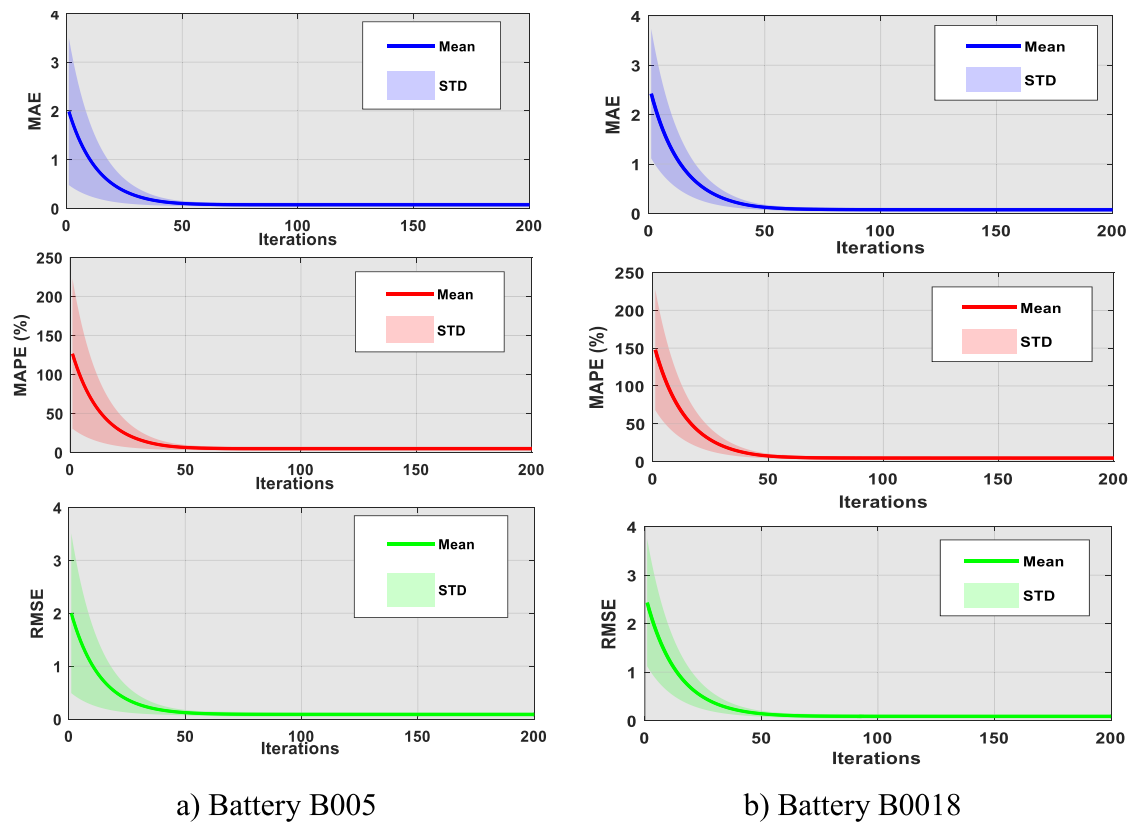


Fig. 8. Convergence behaviour of GCRA during fractional-order optimization.

Table 8

Stability statistics over 20 independent GCRA runs.

Battery	MAE (mean $\pm$ std)	MAPE (mean $\pm$ std)	RMSE (mean $\pm$ std)
B0005	0.0181 $\pm$ 0.0003	0.529 $\pm$ 0.008	0.0222 $\pm$ 0.0004
B0018	0.0091 $\pm$ 0.0002	0.607 $\pm$ 0.007	0.0121 $\pm$ 0.0003

forecasting knee-point behaviour and enhancing practical reliability for battery health management applications.

Table 9

Performance within ± 15 cycles of the knee-point for batteries B0005 and B0018.

Model	B0005		B0018	
	MAE	MAPE (%)	MAE	MAPE (%)
AWFRAGM	0.0078	0.92	0.0091	1.05
EWFRAGM	0.0116	1.38	0.0128	1.49
NDGM-EnKF	0.0139	1.67	0.0154	1.82
WOA-BPNN	0.0164	1.95	0.0182	2.14

3.3.6. Physical consistency

The SOH trajectories estimated by the proposed AWFRAGM model exhibit a physically consistent degradation pattern across the entire lifecycle of both batteries. As shown in Figs. 3d and 4d, the estimated SOH follows a monotonic decreasing trend, in agreement with the irreversible nature of LIB capacity fade (C. Li et al., 2025), and does not display non-physical oscillations or numerical artefacts. The estimated degradation process evolves smoothly from the early-life quasi-linear region to the accelerated degradation regime observed in later cycles. This transition is captured without abrupt discontinuities or instability at the train-test boundary, indicating that the model preserves temporal continuity and does not rely on overfitting to the training data.

Furthermore, the estimated SOH trajectories intersect the predefined EOL threshold of 70 % nominal capacity at cycle indices consistent with the measured degradation data. This alignment confirms that the model-generated SOH estimates remain coherent with established battery ageing criteria and practical BMS definitions of battery failure. Overall, the proposed AWFRAGM produces numerically stable and physically plausible SOH estimates that conform to known LIB degradation mechanisms, supporting its suitability for practical prognostic applications.

3.4. Further tests and analysis

To further evaluate the robustness and statistical reliability of the forecasting results, formal hypothesis-based predictive accuracy tests are conducted. In addition to pointwise error metrics, the DM test and the Superior Predictive Ability (SPA) test are employed to assess whether the observed performance differences between the proposed AWFRAGM model and the benchmarks are statistically significant on the testing horizon.

3.4.1. DM test

The DM test (Zhou et al., 2021) is applied to perform pairwise comparisons between the proposed AWFRAGM model and each benchmark, using the testing datasets of batteries B0005 and B0018. The test is implemented based on the loss differential series constructed from the MAPE, which is consistent with the primary evaluation criterion adopted in this study.

For each battery dataset, the null hypothesis assumes equal predictive accuracy between AWFRAGM and the benchmarks, while the alternative hypothesis indicates superior predictive performance of AWFRAGM. The resulting DM statistics and associated p-values are summarised in Tables 10 and 11 for batteries B0005 and B0018, respectively.

Across both datasets, the computed DM statistics are predominantly positive and exceed the corresponding critical values at the 5 % significance level when comparing AWFRAGM with EWFRAGM and WOA-BPNN. In comparisons with NDGM-EnKF, the DM statistics remain statistically significant at the 5 % level in most cases, with relatively smaller magnitudes reflecting the closer predictive performance between these two grey-based models. These results confirm the presence of statistically significant differences in predictive accuracy between AWFRAGM and the benchmarks over the testing horizon.

Table 10

Diebold-Mariano test results for battery B0005 (using the testing set). Pairwise comparison of the AWFRAGM model against benchmark alternatives.

Pairwise comparison	DM statistic	p-value	Relative MAE reduction (%)	Relative MAPE reduction (%)
NDGM-EnKF	2.18**	0.029	25.0	73.8
EWFRAGM	3.47***	0.001	37.9	83.0
WOA-BPNN	4.92***	< 0.001	56.1	90.7

** and *** indicate statistical significance at the 5 % and 1 % threshold respectively

Table 11

Diebold-Mariano test results for battery B0018 (using the testing set). Pairwise comparison of the AWFRAGM model against benchmark alternatives.

Pairwise comparison	DM statistic	p-value	Relative MAE reduction (%)	Relative MAPE reduction (%)
NDGM-EnKF	2.41**	0.016	59.1	71.0
EWFRAGM	3.89***	< 0.001	66.7	84.2
WOA-BPNN	5.36***	< 0.001	76.9	88.4

** and *** indicate statistical significance at the 5 % and 1 % threshold respectively

3.4.2. SPA test

To complement the pairwise DM analysis and to control for data-snooping effects arising from multiple model comparisons, the SPA test proposed by Hansen (2005) is employed. The SPA test evaluates whether any competing model exhibits statistically superior predictive performance relative to a designated benchmark when all models are assessed simultaneously.

In this study, the proposed AWFRAGM is specified as the benchmark model, while NDGM-EnKF, EWFRAGM, and WOA-BPNN are treated as alternative forecasting models. The loss function is defined using the MAPE-based forecast errors computed over the testing horizon for each battery dataset. A stationary bootstrap procedure with 10,000 replications is adopted to approximate the empirical distribution of the SPA test statistic and to ensure robustness against serial dependence in the loss differentials.

The empirical bootstrap distributions of the SPA test statistics are presented in Fig. 9 for the B0005 dataset and Fig. 10 for the B0018 dataset. In both cases, the SPA test fails to reject the null hypothesis of superior predictive ability of the proposed AWFRAGM at the 5 % significance level. Specifically, for the B0005 battery, the observed SPA statistic is 1.38, with an associated p-value indicating statistical significance at the 1 % level. Similarly, for the B0018 battery, the SPA statistic attains a value of 1.21, which is also statistically significant at the 1 % level. Moreover, across all competing models, the maximum loss differentials relative to AWFRAGM remain non-positive, and the corresponding SPA p-values exceed the selected significance threshold, providing further evidence of the robust predictive superiority of the proposed model.

4. Discussions

The experimental results presented in Section 3 demonstrate that the proposed AWFRAGM consistently achieves superior predictive accuracy for LIB SOH estimation compared to established benchmarks. This Section interprets these findings, elucidates the underlying mechanisms driving the model's performance, and addresses broader implications for battery health prognostics.

4.1. Synthesis of empirical evidence

The experimental results decisively confirm the study's primary objective: the developed AWFRAGM successfully reinforces the NIP principle, yielding superior SOH prediction, particularly during accelerated late-life degradation. The direct comparison with its equally-weighted variant, EWFRAGM, provides the most compelling evidence. AWFRAGM's statistically significant lower errors, such as a 0.53 % versus 3.12 % test MAPE for B0005, demonstrate that the adaptive weighting scheme is not ancillary but fundamental. This confirms that merely using FRAGO is insufficient; the optimization itself must be recency-aware, forcing parameter estimation to prioritize the most recent and relevant dynamics.

Further validation stems from the model's accurate, physically consistent tracking of the full degradation trajectory. The synergistic design of FRAGO is key to this success. The reverse operator inherently

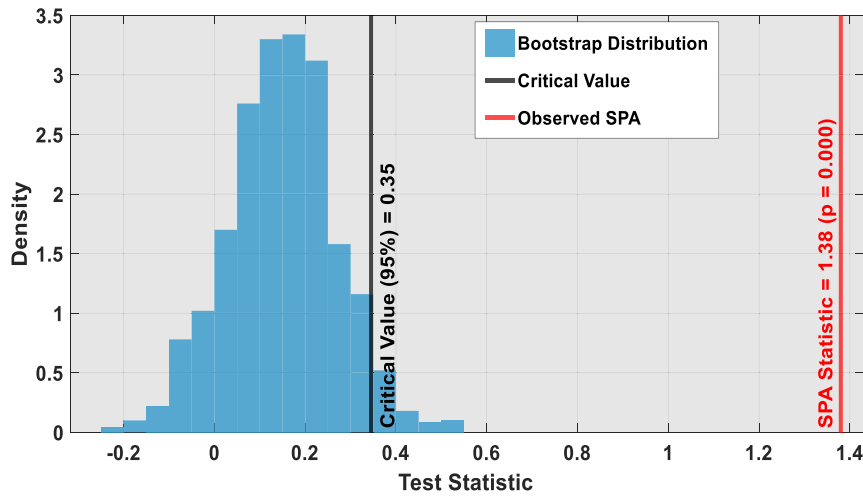


Fig. 9. SPA test for battery B0005. Empirical distribution of the test statistic from 10,000 stationary bootstrap replications. The vertical red line represents the observed SPA statistic, and the shaded blue area corresponds to the critical region at the 5 % significance level, indicating the model's comparative dominance.

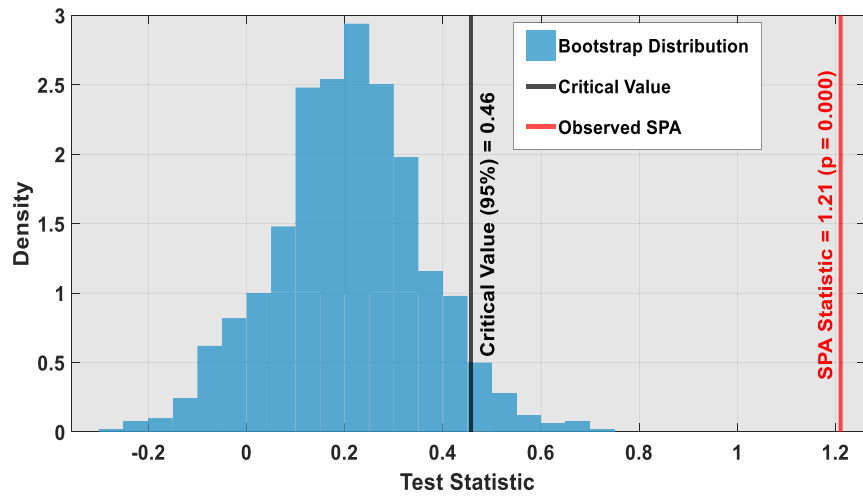


Fig. 10. SPA test for battery B0018. The distribution of bootstrapped SPA test statistics confirms the dominance of the proposed AWFRAGM model. The observed statistic falls within a region rejecting the null hypothesis at high confidence, demonstrating that no competing model shows statistically superior predictive performance.

amplifies new data, while the optimized fractional-order of 0.85 provides an optimal balance, filtering noise without losing sensitivity to the non-linear transition near the knee-point. This integrated, NIP-governed approach also explains AWFRAGM's advantage over other adaptive models like NDGM-EnKF and its robustness compared to the data-hungry WOA-BPNN. The results collectively affirm that the proposed framework effectively addresses the core limitation of conventional GMs by making the prioritization of new information both explicit and operational throughout the modelling process.

4.2. Interpretation of model mechanisms and advantages

The superior performance of AWFRAGM can be understood as a consequence of its coherent architectural design, which embeds the NIP principle at multiple, interdependent levels of the modeling process. This multi-layered reinforcement of recency creates a synergistic forecasting mechanism that is fundamentally aligned with the non-stationary nature of battery degradation. The first and most explicit layer is the adaptive cycle-dependent weighting scheme. By calculating weights Ω_q that increase monotonically with the cycle index q , the model formally declares that recent observations possess greater infor-

mational value for forecasting. This is not merely a post-hoc adjustment but is integrated directly into the model's optimization objective, ensuring that the fitting process is penalized more heavily for errors in the latest stages of the observed degradation. Consequently, parameter estimation is inherently biased toward capturing the most recent dynamics, which are most indicative of imminent changes in the degradation rate, such as the approach to the knee-point region.

The second layer is operationalized through the FRAGO. Traditional forward accumulation, while a cornerstone of GST, inherently dilutes the influence of newer data points within the accumulated sequence. The reverse accumulation operator inverts this dynamic, structurally amplifying the contribution of recent observations during the construction of the modeling sequence $y^{(\xi)}(q)$. When combined with a fractional-order ξ , this operator gains a continuous tuning parameter that controls the memory length and the smoothness of the transformation. A lower ξ places relatively more emphasis on the most recent raw data points, enhancing sensitivity to abrupt changes, while a higher ξ creates a smoother, more averaged historical view. The optimization of ξ via GCRA thus becomes the process of finding the optimal balance between responsiveness to new trends and stability against noise, a balance that is critically informed by the weighting scheme in the first

layer.

This leads to the third layer with the recency-aware metaheuristic optimization. The GCRA does not search for ξ by minimizing a simple sum of squared errors. Instead, it minimizes a weighted error function where the penalties are scaled by the same adaptive weights Ω_q . This creates a feedback loop in which the weights guide the search for an optimal ξ , and the resulting ξ -order reverse accumulation produces a sequence that best fits the weighted data. This closed-loop alignment ensures that the model's structural flexibility (through ξ) is explicitly harnessed to improve fit where it matters most (in the recent data). The final forecasting layer, governed by the whitening equation's time-response function, then propagates this recency-calibrated model state into future predictions. The inverse accumulation operation required to revert to the original SOH scale preserves this temporal fidelity, resulting in forecasts that naturally extend the most recent degradation trajectory without being unduly anchored to early-life behavior.

The collective advantage of this design is a model that maintains the parsimony and computational efficiency of a univariate GM while acquiring a dynamic adaptive focus. It avoids the historical inertia that can plague simpler models when faced with a regime shift like a knee-point. The consistently narrow residual distributions are a direct manifestation of this capability. Unlike benchmarks whose errors tend to diverge as they project further into the accelerated degradation regime, the AWFRAGM's multi-layered NIP mechanism allows it to continuously re-anchor its predictions to the prevailing degradation rate. This architectural coherence between the model's theoretical foundation (NIP), its mathematical operators (FRAGO), and its parameter training strategy (weighted optimization) is the fundamental source of its robust and accurate performance in predicting the complex, non-linear SOH of LIBs.

4.3. Implications for battery management and prognostics

The demonstrated performance of AWFRAGM carries significant practical implications for the design and operation of next-generation BMS. Its ability to accurately track SOH translates directly into enhanced prognostic reliability. By providing high-fidelity SOH estimates with low error persistence, AWFRAGM enables more confident predictions of RUL. This is achieved through the straightforward projection of the model's forecast trajectory to a predefined EOL threshold. The accuracy of this projection is fundamentally improved by the model's inherent sensitivity to the most recent degradation rate, which is the primary determinant of a battery's immediate future.

This capability fosters a shift from reactive to proactive battery management. Early and reliable detection of the transition into accelerated degradation allows a BMS to implement risk-mitigating control strategies in a timely manner. These could include derating charge and discharge currents, adjusting voltage limits, or triggering maintenance alerts, thereby enhancing operational safety and potentially averting catastrophic failures. Furthermore, AWFRAGM's accurate endpoint prediction is invaluable for lifecycle management and economic planning. It provides a reliable basis for scheduling battery replacement in EVs or for assessing the residual value and suitability of a battery pack for secondary applications in stationary energy storage. The model's parsimonious, univariate nature, requiring only historical capacity fade data, is a key practical advantage that simplifies sensor requirements and eases implementation within the computational constraints of embedded BMS hardware, making this advanced prognostic capability accessible for real-world deployment.

From a practical deployment perspective, the computational burden of the proposed AWFRAGM framework can be separated into two stages: offline parameter optimization and online prediction. The GCRA optimization stage has a complexity of approximately $\mathcal{O}(Pop \times Iter_{max} \times N)$, where N denotes the training length. Given the adopted settings ($Pop = 50$, $Iter_{max} = 200$, $N = 93$), the total number of fitness evaluations remains moderate and is executed only during initialization or periodic recalibration rather than at every prediction step.

Once the optimal parameters are identified, the online SOH prediction reduces to recursive grey model computations with linear complexity $\mathcal{O}(N)$, involving only basic arithmetic and fractional accumulation operations. Therefore, the computational cost of a single prediction is very low and compatible with rolling-window implementation in embedded BMS hardware. In practice, parameter updates can be triggered periodically or when residual thresholds indicate regime shifts, further limiting computational demand. Compared with data-intensive neural-network approaches requiring continuous retraining, the proposed model maintains a compact parameter structure and minimal memory requirements, supporting its feasibility for real-time BMS applications.

4.4. Limitations and future research directions

While the results presented herein are promising, the validation scope of this study inherently defines its primary limitation. The model was evaluated using two well-known laboratory datasets under controlled conditions. Its performance must therefore be rigorously assessed across a wider spectrum of battery chemistries, such as lithium iron phosphate (Wang et al., 2026) and various nickel-manganese-cobalt formulations (Evro et al., 2024), and under realistic, variable operating profiles that include dynamic loads, diverse thermal environments, and irregular usage patterns (Sapnken et al., 2025). Such testing is essential to confirm the robustness and generalizability of the proposed approach outside the benchmark setting.

Nevertheless, it is important to emphasise that the proposed framework is not structurally tailored to a specific degradation curve shape or chemistry. The fractional-order mechanism provides adjustable memory depth, allowing the model to flexibly adapt to prolonged plateau behaviours characteristic of certain chemistries (like in LFP) or to earlier and more progressive fade patterns often observed in NMC systems. Simultaneously, the NIP-based adaptive weighting enhances sensitivity to regime shifts, enabling the model to respond dynamically whether degradation acceleration occurs early or late in the life cycle. In this sense, the theoretical foundations of the framework are chemistry-agnostic, and future cross-chemistry validation represents an extension of scope rather than a structural modification of the model.

Furthermore, the study's motivation centred on the critical challenge of knee-point prediction, yet the evaluation metrics focused on overall SOH accuracy. A more direct assessment in future work would involve quantifying the model's performance in explicitly identifying the onset cycle of the knee-point itself, providing a targeted metric for its core prognostic claim. To transition from a batch analysis tool to an online BMS component, the development of an incremental or rolling-window implementation is a necessary next step. This would allow the model to continuously assimilate new data, recompute adaptive weights, and update its parameters in real-time, embodying a true metabolic grey forecasting system.

Finally, the model's current univariate framework could be expanded. The adaptive weighting concept could be refined to reflect not just temporal recency but also the severity of operational stressors experienced in each cycle. Exploring a multivariate extension that incorporates correlated signals like internal resistance or temperature swing as explanatory variables, presents a compelling avenue for potentially enhancing predictive accuracy and capturing a more holistic view of the degradation state (Moyo and Mwaniki, 2024).

5. Conclusion

Accurate prediction of LIB SOH remains a formidable challenge due to the nonlinear and non-stationary nature of degradation, particularly the abrupt acceleration associated with the knee-point phenomenon. This study addressed a fundamental limitation of conventional grey models: the insufficient and non-systematic prioritization of recent degradation information. To overcome this structural weakness, a novel

adaptively weighted fractional-order reverse accumulation grey model (AWFRAGM) was developed. The core innovation of this work lies in the systematic and multi-layer reinforcement of the New Information Priority (NIP) principle across three coordinated stages: adaptive cycle-dependent weighting during sequence construction, fractional-order reverse accumulation for enhanced recency-sensitive memory representation, and recency-aware metaheuristic parameter optimization. This integrated architecture transforms NIP from a conceptual guideline into an operational modelling mechanism.

Comprehensive validation on NASA benchmark datasets demonstrated that AWFRAGM yields low prediction errors compared to grey and neural-network-based benchmarks. The model achieved testing MAPEs below 1 %, maintained physically consistent monotonic SOH trajectories, and showed statistically significant superiority under DM and SPA tests. These results confirm that reinforcing temporal recency at structural and optimization levels substantially improves late-life degradation tracking.

Beyond predictive accuracy, the practical viability of AWFRAGM for embedded Battery Management Systems (BMS) was evaluated from an implementation perspective. Owing to its parsimonious structure, the model requires only historical capacity data and involves low-dimensional parameter estimation. For a single prediction cycle with fixed parameters, the computational complexity is linear in the window size, consisting primarily of fractional accumulation and closed-form grey model evaluation. The metaheuristic optimization (GCRA) is required only during parameter update stages and operates over a low-dimensional search space (primarily the fractional order ξ), resulting in limited computational overhead compared with deep learning frameworks.

A feasible online implementation framework can be structured using a rolling-window mechanism: model parameters, including ξ , are re-optimized every N cycle using the most recent windowed data, while fixed-parameter forecasting is performed between update intervals. Update triggers may be scheduled periodically or activated adaptively when residual errors exceed predefined thresholds. Given the small parameter space and modest iteration requirements of GCRA, the optimization stage can be executed within acceptable time budgets for modern embedded BMS microcontrollers, particularly since updates are infrequent relative to measurement sampling rates. Integration into conventional BMS architectures would occur at the prognostics layer, interfacing directly with capacity estimation modules without modifying lower-level sensing or state-estimation routines.

The present study is limited by validation on laboratory-grade datasets under controlled conditions. Future work should expand evaluation across diverse chemistries and real-world duty cycles, incorporate explicit knee-point detection metrics and uncertainty quantification, and further benchmark embedded runtime performance on representative BMS hardware platforms. In all, this work advances grey system theory for battery prognostics by establishing a structurally integrated, recency-driven modelling paradigm that combines theoretical rigor with practical feasibility. The proposed AWFRAGM not only enhances predictive accuracy but also provides a technically viable pathway toward real-time deployment in next-generation intelligent BMS.

CRedit authorship contribution statement

Naiming Xie: Writing – review & editing, Supervision, Software, Investigation, Formal analysis, Data curation. **Yong Wang:** Writing – review & editing, Visualization, Validation, Software, Resources, Investigation. **Flavian Emmanuel Sapnken:** Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Reagan Jean Jacques Molu:** Writing – review & editing, Visualization, Investigation, Formal analysis, Conceptualization. **Ghislain Junior Bangoup Ntegm:** Writing – review & editing, Validation, Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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